

Subject Description Form

Subject Code	AF4232
Subject Title	FinTech in Banking and Capital Markets
Credit Value	3
Level	4
Normal Duration	One Semester
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This course is designed to introduce the FinTech and its applications in the banking industry and capital markets. By studying the conceptual ideas of the FinTech applications as well as the relevant cases, the students are able to learn the difference between traditional and digital banking, and apply theories and knowledge to conduct analysis on digital banking services and products, such as mobile and internal banking, robo-advisory model, electronic know your customer (eKYC) process, Blockchain, open application programming interface (Open API), regulatory technology (RegTech) etc. The digital capital markets would be further discussed, and the students would learn the concept of decentralized finance (DeFi) and be able to understand the digital assets such as Bitcoin, Ethereum, exchange token and others, and create the smart contracts accordingly. Regulations on digital banking and capital markets would be discussed in class. This course helps achieve the BBA (Hons) in Digital Finance, Banking and Investment outcomes by enabling students to better understand the analytical tools in accounting and finance (Outcomes 11a and 11c), and help them to better apply the technology and data analytics skills to solve accounting and finance problems faced in real-life situations in an ethical manner (Outcomes 11a and 11c).
Intended Learning Outcomes <i>(Note 1)</i>	Upon successful completion of this subject, students should be able to: a. Develop contextualised understanding of FinTech and its applications in the Banking Industry and Capital markets. b. Understand the development of Virtual Banks in HK and overseas c. Understand the anatomy of digital assets and capital markets (coins, tokens, futures, ETF, stablecoin, De-Fi, NFT); d. Understand challenges of FinTech in protecting personal data privacy, and complying with various regulatory requirements.

Subject Synopsis/ Indicative Syllabus <i>(Note 2)</i>	FinTech in Banking Industry An overview of the FinTech applications in banking industry and products: to understand the genesis of virtual banks and virtual insurers, and how they differ from traditional banks and insurers; to understand the impact of digitalization on the bank’s front, middle and back office functions and the insurer’s value chain. Digital Banking Services and Products Introduce various digital banking services and products such as mobile and internal banking, robo-advisory model, electronic know your customer (eKYC) process, Blockchain, open application programming interface (Open API), regulatory technology (RegTech) etc. Learn and think critically the formation process of FinTech business strategies, solutions and implementation to resolve problems in the Banking Industry. FinTech in Capital Markets Introduce the concept of digital assets, and digital capital markets, such as Bitcoin, Ethereum, exchange token, De-Fi, stablecoin, non-fungible token, Bitcoin futures, Bitcoin ETF, crypto index fund, etc. Regulations & Ethical Issues Understand challenges of FinTech in protecting personal data privacy, and complying with various regulatory requirements, regulatory ambiguity over digital assets and digital capital markets, other ethical issues with financial technologies					
Teaching/Learning Methodology <i>(Note 3)</i>	Key concepts and techniques will be introduced through lectures. Cases will be used to enable students to understand the various applications of Fintech in banking and financial markets. As a practical subject, a lot of emphases will be placed on project work. Students will be required to deliver a project with a focus on real-world accounting and finance issues. By completing the project, students should have hands-on experience in using the knowledge they have learnt in class to solve accounting and finance problems in practice. Students are also encouraged to share their views and experiences actively with their lecturer and classmates.					
Assessment Methods in Alignment with Intended Learning Outcomes <i>(Note 4)</i>	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed			
			a	b	c	d
	1. Class participation	5%	√	√	√	√
	2. Quizzes & assignments	15%	√	√	√	√
	3. Group project	30%	√	√	√	√
	4. Final examination	50%	√	√	√	√
	Total	100 %				

	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Class participation – Students have to read assigned reading materials, complete exercises and participate actively in class discussion. They will be assessed on their understanding of the key concepts and techniques learnt in the subject. Quizzes & Assignments – Quizzes & assignments are used to test students’ understanding of the teaching materials and achievement of the intended learning outcomes through a more in-depth investigation of different topics. Group Project – Group project is used for students to apply what is taught in class and allow the students to learn from one another. Final Examination – Final examination is used to test students’ overall ability in applying the knowledge learnt in the subject. <i>Note: The specific requirements on individual assessment components discussed above could be adjusted based on the pedagogical needs of subject lecturers.</i>	
Student Study Effort Expected	Class contact:	
	▪ Seminars	39 Hours
	Other student study effort:	
	▪ Self-study including reading textbook and preparation for quizzes, assignments and examination	55 Hours
	▪ Preparation for group project	20 Hours
	Total student study effort	114 Hours
Reading List and References	<i>Authorization of Virtual Banks A Guideline issued by the Monetary Authority under Section 16(10), Hong Kong Monetary Authority</i> https://www.hkma.gov.hk/media/eng/doc/key-information/guidelines-and-circular/guideline/guideline_eng_virtual_bank_20180608.pdf) <i>DeFi and the Future of Finance</i>, 1st Edition, Campbell Harvey, Ashwin Ramachandran and Joey Santoro, Wiley <i>e-HKD: A technical perspective</i>, Hong Kong Monetary Authority https://www.hkma.gov.hk/media/eng/doc/key-functions/financial-infrastructure/e-HKD_A_technical_perspective.pdf)	